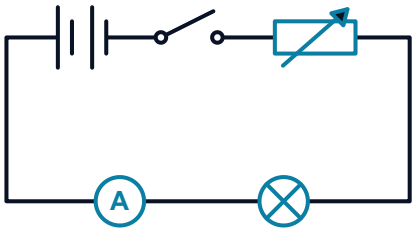
DO NOW · REQUIRED PRACTICAL

# I–V characteristic of a filament lamp

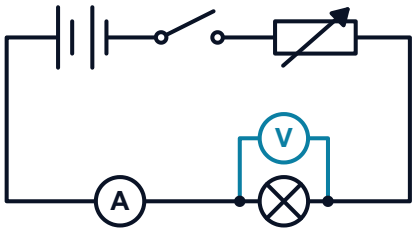
**Electricity**
Paper 1 · Combined and Separate
6 steps

A student finds the I–V characteristic of a filament lamp.

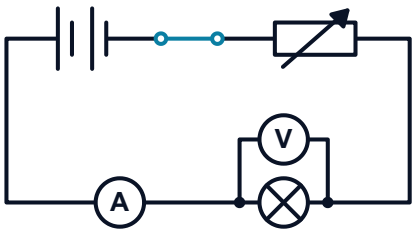
## 1 Connect the lamp in series with a power supply, a variable resistor and an ammeter.




## 2 Connect a voltmeter in parallel with the lamp.

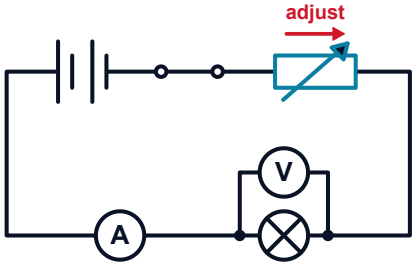



## 3 Record the current and the potential difference.



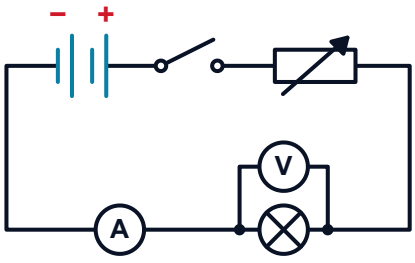

I = ___ A  V = ___ V

## 4 Adjust the variable resistor and record a new pair of readings; do this several times.



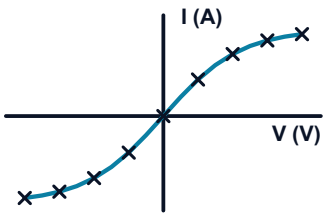

record a new pair of readings; repeat several times

## 5 Reverse the connections to the power supply and take the readings again.




reverse the connections to the supply

## 6 Plot a graph of current against potential difference.




I against V, both directions

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